

Cloud Services for Generative AI

Introduction

Cloud platforms have become the foundation for building, training, deploying, and scaling Generative AI applications. Rather than managing infrastructure manually, organizations can leverage managed AI services that provide foundation models, machine learning platforms, vector databases, document intelligence, speech processing, computer vision, and natural language processing capabilities.

The three major cloud providers—Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP)—offer comprehensive AI ecosystems that enable developers to build production-ready AI applications with minimal operational overhead.

AWS AI Services

Amazon Web Services provides one of the most comprehensive AI ecosystems, covering the complete AI lifecycle from data ingestion to model deployment and monitoring.

Amazon SageMaker

Amazon SageMaker is AWS's end-to-end machine learning platform designed for building, training, deploying, and monitoring machine learning models. It provides managed notebooks, distributed training, model hosting, automated machine learning, and MLOps capabilities.

Common Use Cases

- Training custom machine learning models
- Model deployment and inference
- MLOps pipelines
- Hyperparameter tuning
- Batch inference
- Model monitoring

Amazon Bedrock

Amazon Bedrock is a fully managed Generative AI platform that provides access to multiple foundation models through a single API. It enables developers to build GenAI applications without managing infrastructure.

Key Features

- Access to multiple foundation models
- Serverless model inference
- Knowledge Bases for RAG
- Agent capabilities
- Fine-tuning support
- Guardrails for responsible AI

Common Use Cases

- AI chatbots
- Enterprise assistants
- Document summarization
- Content generation
- AI-powered search
- Agentic AI applications

Amazon OpenSearch Service

Amazon OpenSearch Service supports vector search capabilities, making it an excellent choice for Retrieval-Augmented Generation (RAG) applications.

Key Features

- Vector embeddings
- Semantic search
- Hybrid search
- Similarity search

- Large-scale indexing

Common Use Cases

- Knowledge base search
 - Semantic document retrieval
 - Enterprise search
 - AI assistants
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Amazon Textract

Amazon Textract extracts text, forms, tables, and structured information from scanned documents using OCR and document intelligence.

Common Use Cases

- Invoice processing
 - Medical documents
 - Insurance claims
 - Financial reports
 - Identity verification
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Amazon Comprehend

Amazon Comprehend is AWS's Natural Language Processing service that analyzes text using machine learning.

Features

- Entity recognition
- Sentiment analysis
- Key phrase extraction
- Language detection
- Topic modeling
- PII detection

Amazon Rekognition

Amazon Rekognition provides image and video analysis capabilities.

Features

- Object detection
 - Face recognition
 - Face comparison
 - Text detection
 - Content moderation
 - Image classification
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Amazon Transcribe

Amazon Transcribe converts speech into text using automatic speech recognition.

Use Cases

- Meeting transcription
 - Call center analytics
 - Voice assistants
 - Video subtitles
 - Medical transcription
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Amazon Personalize

Amazon Personalize builds recommendation systems using machine learning.

Applications

- Product recommendations
- Movie recommendations

- Personalized content
 - Customer personalization
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Agent Core

Agent Core enables developers to build intelligent AI agents capable of reasoning, planning, tool usage, and multi-step task execution.

Common Applications

- AI copilots
 - Autonomous workflows
 - Enterprise assistants
 - Customer support agents
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Microsoft Azure AI Services

Microsoft Azure offers a complete AI ecosystem focused on enterprise-grade AI development.

Azure AI Foundry

Azure AI Foundry is the central platform for developing, managing, and deploying AI applications.

Components

- Model catalog
 - Prompt management
 - Evaluation tools
 - Deployment management
 - AI orchestration
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Azure OpenAI Service

Provides managed access to OpenAI models within Azure's secure infrastructure.

Common Applications

- Chatbots
 - AI copilots
 - Code generation
 - Document analysis
 - Content generation
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Azure Machine Learning

Azure Machine Learning provides an end-to-end machine learning platform.

Features

- Training
 - Deployment
 - Model registry
 - AutoML
 - MLOps
 - Monitoring
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Azure AI Vision

Computer vision services for image analysis.

Capabilities

- OCR
- Object detection
- Image captioning
- Face detection
- Image tagging

Azure AI Custom Vision

Enables organizations to build custom computer vision models without extensive ML expertise.

Azure AI Speech Services

Speech-related AI services including:

- Speech-to-text
 - Text-to-speech
 - Speech translation
 - Speaker recognition
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Azure AI Translator

Provides multilingual translation across numerous languages.

Use Cases

- Website localization
 - Document translation
 - Real-time communication
 - Global applications
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Azure AI Document Intelligence

Extracts structured information from business documents.

Applications

- Forms

- Receipts
 - Contracts
 - Invoices
 - Identity documents
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Azure Bot Service

A managed platform for developing conversational AI applications.

Applications

- Customer support
 - Virtual assistants
 - FAQ bots
 - Enterprise bots
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Azure AI Personalizer

Delivers personalized experiences using reinforcement learning.

Azure AI Anomaly Detector

Detects unusual patterns in datasets for predictive analytics and monitoring.

Azure AI Search

Azure AI Search combines traditional search with vector search and semantic capabilities for RAG applications.

Google Cloud AI Services

Google Cloud provides an AI-first platform centered around Vertex AI.

Vertex AI

Vertex AI is Google's unified machine learning platform.

Features

- Model training
 - AutoML
 - Model deployment
 - MLOps
 - Prompt engineering
 - Foundation model integration
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Gemini

Google's family of multimodal foundation models accessible through Vertex AI.

Applications

- Chat assistants
 - Code generation
 - Document understanding
 - Vision-language tasks
 - Multimodal AI
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Vertex AI Search

Enterprise search platform supporting semantic search and Retrieval-Augmented Generation.

Document AI

Processes structured and unstructured documents using AI.

Applications

- Invoice extraction
 - Contract analysis
 - Financial documents
 - Healthcare documents
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Vision AI

Google Cloud's computer vision platform.

Capabilities

- OCR
 - Image labeling
 - Face detection
 - Product search
 - Object tracking
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Speech-to-Text and Text-to-Speech

Provides speech recognition and speech synthesis services.

Natural Language AI

Offers advanced NLP capabilities including:

- Entity recognition
 - Sentiment analysis
 - Syntax analysis
 - Content classification
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AI Service Comparison

AI Capability	AWS	Azure	GCP
Foundation Models	Amazon Bedrock	Azure OpenAI	Gemini
ML Platform	SageMaker	Azure Machine Learning	Vertex AI
Vector Search	OpenSearch	Azure AI Search	Vertex AI Search
OCR & Documents	Textract	Document Intelligence	Document AI
Computer Vision	Rekognition	Azure AI Vision	Vision AI
Speech AI	Transcribe	Azure Speech	Speech-to-Text
NLP	Comprehend	Text Analytics	Natural Language AI

Choosing the Right Cloud Platform

AWS

Best suited for organizations requiring a mature ecosystem with extensive AI services, enterprise scalability, and strong infrastructure integration.

Azure

Ideal for enterprises already using Microsoft technologies and requiring seamless integration with enterprise productivity tools.

Google Cloud

Excellent for organizations focused on advanced AI research, machine learning, data analytics, and Google's Gemini ecosystem.

Module Summary

Modern Generative AI applications rely heavily on managed cloud AI services. AWS, Azure, and Google Cloud each provide comprehensive platforms covering machine learning, foundation models, vector search, document intelligence, computer vision, speech processing, and natural language understanding. Selecting the right platform depends on existing infrastructure, business requirements, scalability needs, and preferred AI ecosystem.

Module 17: Deployment Strategies

Introduction

Building an AI application is only one part of the development lifecycle. Deploying, scaling, monitoring, and maintaining AI systems in production requires selecting an appropriate deployment strategy based on application architecture, scalability requirements, operational complexity, and infrastructure management preferences.

Cloud providers offer multiple deployment models, ranging from fully serverless architectures to managed containers, Kubernetes platforms, and virtual machines.

Deployment Categories

Deployment strategies generally fall into the following categories:

- Serverless Computing
 - Container-Based Deployment
 - Kubernetes
 - Virtual Machines
 - Frontend Hosting
 - Platform as a Service (PaaS)
 - Workflow Automation
 - Event-Driven Architecture
 - CI/CD Pipelines
 - Machine Learning Platforms
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AWS Deployment Strategies

AWS Lambda

A serverless compute service that automatically scales without requiring server management.

Best For

- REST APIs
- Event processing
- Scheduled jobs
- AI inference endpoints
- Lightweight backend services

Infrastructure Management: None

API Gateway + Lambda

Combines API Gateway with Lambda to build scalable REST APIs.

Best For

- Serverless APIs
 - Mobile backends
 - AI microservices
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Application Load Balancer + Lambda

Routes HTTP requests to Lambda functions for advanced routing scenarios.

Amazon ECS (EC2)

Runs containerized applications on EC2 instances.

Best For

- Long-running services
 - GPU workloads
 - Full infrastructure control
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Amazon ECS Fargate

Runs containers without managing servers.

Best For

- Modern backend applications
 - AI APIs
 - Microservices
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AWS App Runner

Simplifies container deployment with automatic scaling.

Best For

- Fast deployment
 - Web APIs
 - AI services
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Amazon EKS

Managed Kubernetes platform for enterprise-scale deployments.

Best For

- Large microservice architectures
 - AI platforms
 - Enterprise Kubernetes
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Amazon EKS with Fargate

Serverless Kubernetes where AWS manages compute resources.

Amazon EC2

Virtual machines providing complete operating system control.

Best For

- GPU instances
 - Legacy applications
 - Custom environments
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EC2 Auto Scaling Groups

Automatically adjusts EC2 capacity based on demand.

Amazon S3 Static Website Hosting

Hosts static frontend applications.

Best For

- React
 - Angular
 - Vue
 - Static documentation
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S3 + CloudFront

Global frontend hosting with CDN acceleration.

Best For

- Production web applications
 - Low-latency delivery
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Elastic Beanstalk

Platform as a Service for deploying web applications with minimal operational effort.

AWS Step Functions

Coordinates multiple AWS services into reliable workflows.

Common Applications

- AI pipelines
 - Document processing
 - Data workflows
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Amazon EventBridge

Event-driven architecture platform.

Use Cases

- Microservices
 - Automation
 - Event processing
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AWS CodePipeline

Managed CI/CD pipeline service.

GitHub Actions

Popular cloud-native CI/CD platform supporting automated build, testing, and deployment workflows.

SageMaker Deployment

Purpose-built environment for deploying machine learning models into production.

AWS Core Cloud Building Blocks

Compute

- Amazon EC2
- AWS Lambda
- Amazon ECS
- Amazon EKS

Storage

- Amazon S3
- Amazon EBS

Databases

- Amazon DynamoDB
- Amazon RDS
- Amazon OpenSearch
- Amazon Redshift

AI

- Amazon Bedrock
- Amazon SageMaker
- Amazon Q

Networking

- Amazon CloudFront
- Elastic Load Balancer
- Amazon VPC

Security

- AWS Identity and Access Management (IAM)
 - IAM Users
 - IAM Roles
 - IAM Policies
 - Amazon Cognito
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Microsoft Azure Deployment Strategies

Azure provides equivalent deployment options across serverless, containers, Kubernetes, VMs, frontend hosting, workflows, and CI/CD.

Key Services

- Azure Functions
 - API Management
 - Application Gateway
 - Azure Container Apps
 - Azure App Service
 - Azure Kubernetes Service (AKS)
 - AKS Virtual Nodes
 - Azure Virtual Machines
 - VM Scale Sets
 - Azure Static Web Apps
 - Azure Blob Storage + CDN
 - Logic Apps
 - Event Grid
 - Azure DevOps Pipelines
 - GitHub Actions
 - Azure Machine Learning
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Google Cloud Deployment Strategies

Google Cloud offers modern deployment services optimized for cloud-native applications.

Key Services

- Cloud Functions
- API Gateway
- Cloud Load Balancer
- Cloud Run
- Google Kubernetes Engine (GKE)
- GKE Autopilot
- Compute Engine
- Managed Instance Groups
- Firebase Hosting
- Cloud Storage + Cloud CDN
- App Engine
- Workflows
- Eventarc
- Vertex AI

Cross-Cloud Deployment Comparison

Deployment Pattern	AWS	Azure	Google Cloud
Serverless Compute	Lambda	Azure Functions	Cloud Functions
API Gateway	API Gateway	API Management	API Gateway
Containers	ECS Fargate	Container Apps	Cloud Run
Managed App Platform	App Runner	App Service	App Engine

Kubernetes	EKS	AKS	GKE
Virtual Machines	EC2	Azure VM	Compute Engine
Auto Scaling	Auto Scaling Groups	VM Scale Sets	Managed Instance Groups
Static Frontend	S3 + CloudFront	Blob Storage + CDN	Cloud Storage + CDN
Workflow Orchestration	Step Functions	Logic Apps	Workflows
Event Processing	EventBridge	Event Grid	Eventarc
Machine Learning Platform	SageMaker	Azure Machine Learning	Vertex AI

Selecting the Right Deployment Strategy

The appropriate deployment model depends on the application's architecture and operational requirements:

- **Serverless** is ideal for event-driven APIs and lightweight workloads with minimal infrastructure management.
 - **Containers** provide portability and flexibility for modern microservices.
 - **Kubernetes** is best suited for large-scale enterprise applications requiring orchestration and advanced scalability.
 - **Virtual Machines** are appropriate when full operating system control or specialized hardware such as GPUs is required.
 - **Static hosting with CDN** delivers high-performance frontend applications globally.
 - **Workflow and event services** simplify orchestration and automation of distributed systems.
 - **CI/CD platforms** automate software delivery and reduce deployment risk.
 - **Managed ML platforms** streamline the production deployment of machine learning models.
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Conclusion

Modern cloud platforms provide complete ecosystems for developing and deploying Generative AI applications. AWS, Azure, and Google Cloud each offer mature AI services, deployment models, workflow automation, and machine learning platforms. Understanding the strengths of each service enables architects and developers to choose the right technologies for scalable, secure, and production-ready AI systems. As organizations increasingly adopt cloud-native AI architectures, proficiency in these platforms has become a critical skill for AI engineers, machine learning practitioners, and cloud architects.